Characterization of µm-sized sample surfaces by X-ray reflectivity

Steffen Tober[1], Wieland Corts[1,2], Steven Leake[3], Toka Matar[1,2], Lisa Randolph[1], Breno Rabelo Couthino Saraiva[4,5], Vedran Vonk[4], Andreas Stierle[4,5], Hans-Georg Steinrück[1,2]

[1]Forschungszentrum Jülich GmbH, Institute for a Sustainable Hydrogen Economy (INW), Jülich/Germany

[2]RWTH Aachen University, Institute of Physical Chemistry, Aachen/Germany

[3]ESRF – The European Synchrotron, Grenoble/France,

[4] Deutsches Elektronen-Synchrotron (DESY), Centre for X-Ray and Nanoscience (CXNS), Hamburg/Germany

[5]University of Hamburg, Physics Department, Hamburg/Germany

X-ray reflectivity (XRR) is an established method for non-destructive probing of electron density profiles at surfaces and interfaces with sub-nm-resolution. Regularly applied to semiconductors [1], polymer thin-films [2] or model catalysts [3], XRR is commonly limited to flat, mm-sized samples due to the large effective footprint of the X-ray beam at relatively small incident angles. The resulting beam spillover significantly reduces the reflected signal and increases the background for smaller samples, rendering XRR of µm- and nm-sized samples to date infeasible.

To study surfaces and interfaces of individual particles in multigrain catalysts and for dispersed catalyst particles, a significant reduction of the footprint is necessary to increase the signal to noise ratio and to prevent the simultaneous illumination of multiple crystallites. Toward this end and inspired by the pioneering work by Stubbs *et al.* [4] to extend crystal truncation rod (CTR) measurements to µm-sized crystals, we developed a methodology for (low-angle) XRR of µm-sized samples by employing the nm-sized X-ray beams available at modern synchrotron X-ray sources.

In this contribution, we present proof-of-principle XRR experiments using the nm-sized X-ray beam of ESRF ID01 [5] to measure XRR of test structures in the µm-range with reduced spillover. Specifically, we compared the XRR of a 700 µm “standard/baseline” large sample to the XRR of a “small” 30 µm sample using two

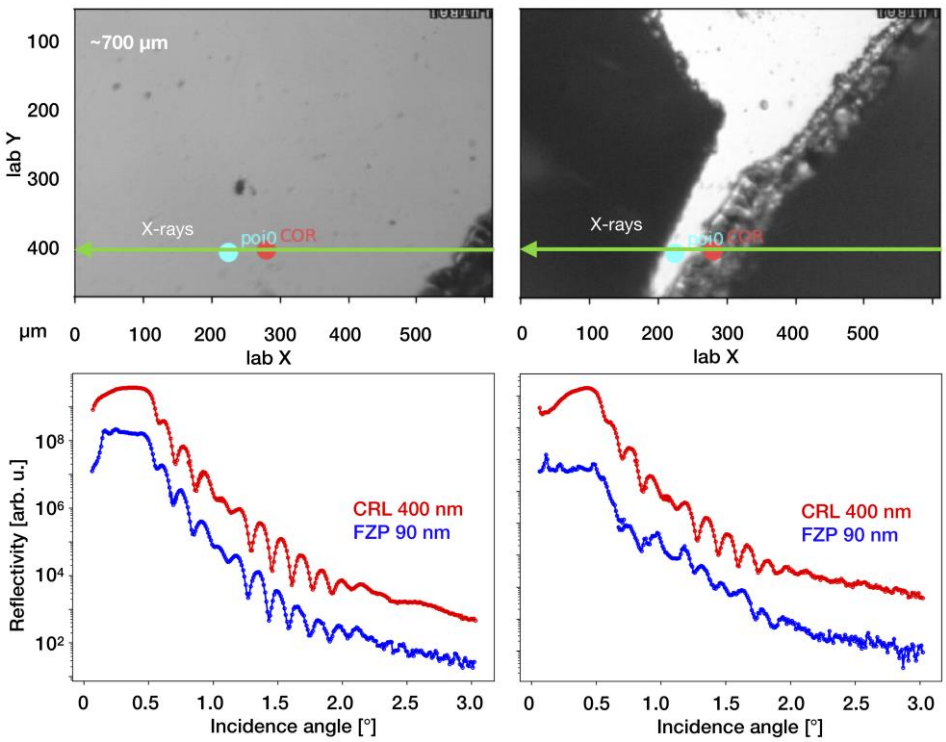

Fig. 1: Microscopy images (*top*) and XRR curves (*bottom*) of an Au(20 nm)/Ti(5 nm)/Si sample (edge of cleaved wafer). The left sample mimics a "standard/baseline" large sample (700 µm) and the left "tip" sample mimics a "small" sample (30 µm). XRR measurements were done with a 400 nm beam focussed with compound refractive lenses (CRL) and a 90 nm beam focussed by a Fresnel zone plate (FZP).

beam sizes each (i.e., 400 nm beam focused with compound refractive lenses (CRL) and a 90 nm beam focused by a Fresnel zone plate (FZP)). First results are presented in Fig. 1 showing reasonable agreement between the XRR of the baseline and small sample; the fringes result from interferences of X-rays reflected at the various interfaces present in the sample. They represent the Fourier transform of the sample's electron density profile. This indicates the general feasibility of XRR of µm-sized samples using nm-sized beams thereby reducing the footprint. Adjustments to the scan procedure allowed a reliable, precise positioning of the beam on the sample and will enable adapting our method for studies of catalysts under *in situ* and *operando* conditions.

[1] S. L. Moffitt *et al.,* JPCC **127** 23099 (2023) [2] P. Lakner *et al.,* Langmuir **36** 13448 (2020) [3] R. Gleißner *et al.*, ACS Appl. Nano Mater. **6** 8004 (2023) [4] J. Stubbs *et al.*, Clays Clay. Miner. **69**, 688 (2021) [5] S. J. Leake *et al.,* J. Synchrotron Rad. **26** 571 (2019)